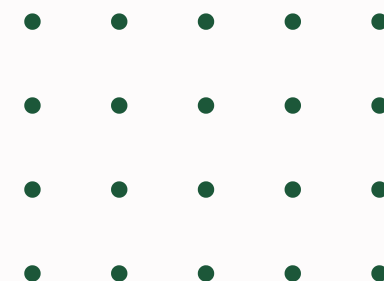


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Multi-Layered Fraud, Waste, and
Abuse (FWA) Detection Engine &
SIU Investigator Workspace

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Executive Summary

The Problem

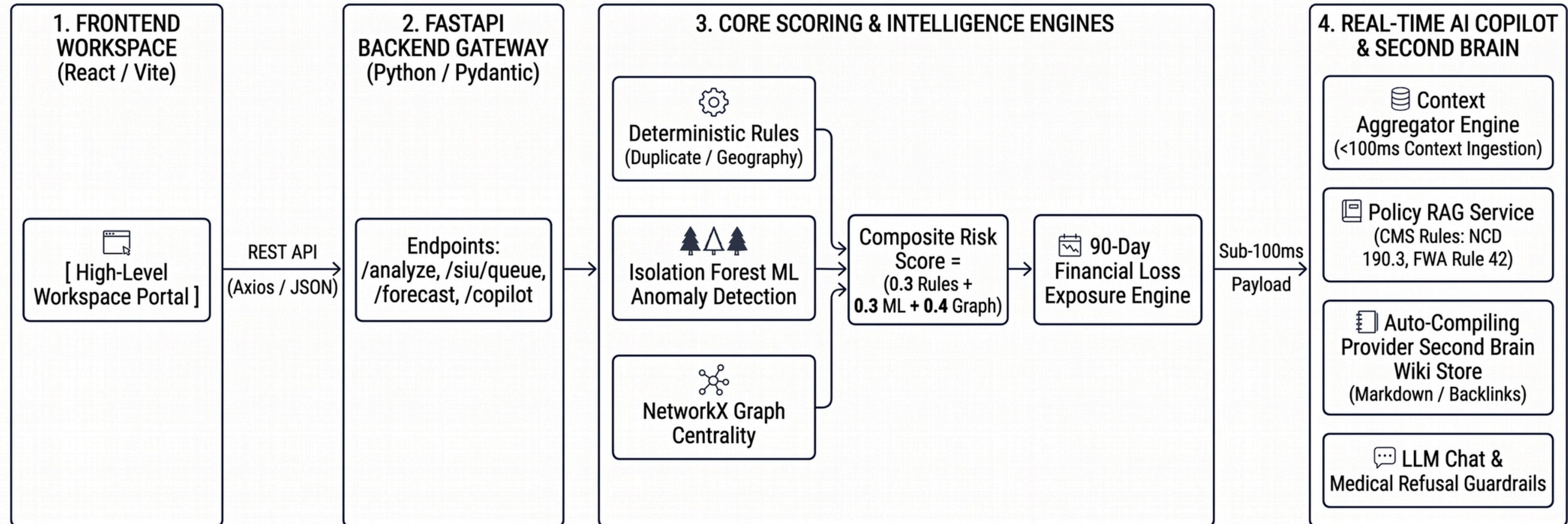
- **Massive Financial Loss:** Healthcare FWA costs insurance providers billions annually due to undetected billing anomalies.
- **Legacy System Deficiencies:** Rigid rule engines miss complex fraud schemes, while standalone ML models operate as unexplainable "black boxes".
- **Investigator Fatigue & Memory Loss:** SIU teams face severe alert fatigue from isolated alerts and lack persistent memory tools across cases

The Proposed Solution

- **Tri-Factor Composite Risk Scoring:** Replaces isolated alerts by combining Business Rules (30%), Machine Learning (30%), and Network Graph Centrality (40%) into a single prioritized score.
- **Forward Loss Exposure Forecasting:** Calculates daily provider billing velocity to project cumulative 90-day financial loss horizons.
- **AI Copilot & Second Brain Workspace:** Sub-100ms context aggregation, policy RAG citations, refusal guardrails, and auto-compiling provider wiki summaries.



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Synthetic Data & Fraud Injection

- **Zero-HIPAA Data Generator:** Generated 5,000+ synthetic claims, 200 provider NPIs, 1,000 members, and 50 facilities via Faker with zero missing primary keys (test_data_gen.py).
- **Hardcoded Synthetic Fraud Patterns:**
 1. Phantom Billing: Dr. Synthetic-A billing for non-existent patients.
 2. Upcoding Spikes: CPT code manipulation driving billing 4 times above peer baselines.
 3. Impossible Geography: Dr. Synthetic-B submitting claims in Chennai and Delhi within 2 hours.
- **Handoff Output:** Exported claims.json and claims.csv to Backend and Copilot services.

NetworkX Graph Analytics

- **Relationship Mapping: Nodes** = Providers, Members, Clinics; **Edges** = Claims, Referrals, Shared Tax IDs.
- **Coordinated Fraud Ring Analytics:** Connected component and centrality algorithms identifying high-density billing loops.
- **Graph Handoff:** Exported centrality scores (>0.75) to Composite Scoring Engine and GET /api/v1/graph/{case_id} to React Flow UI.

FastAPI Core Gateway

- Configured a modular zero-cost Python backend architecture with rigid Pydantic schemas (Claim, Provider, SIUCase, GraphPayload).
- Verified API endpoints respond with HTTP 200 in sub-20 ms base execution time.

Dual Detection Engine

- **Rules Engine (rules_engine.py):** Checks duplicate billing (<5 mins), impossible geography (<2hrs), and upcoding spikes (>= 4* benchmark).
- **ML Anomaly Engine (ml_engine.py):** Isolation Forest model trained on claim vectors, outputting a normalized anomaly score (0.0 - 1.0).

Master Composite SIU Case Ranking

- Master Composite SIU Case Ranking Matrix (app/services/siu_ranking.py)
- Master Formula:
- Composite Risk Score = (0.4 * Rules) + (0.4 * ML Anomaly) + (0.2 * Graph Centrality)
- Ranks SIU cases dynamically to eliminate alert fatigue and surface highest-exposure cases at the top.

Financial Loss Exposure Forecasting & Context Aggregator

- **90-Day Loss Exposure Forecasting** (`app/services/forecasting.py`)
 - **Daily Billing Velocity Math:** Calculates a provider's historical billing velocity (billing rate/day) rather than relying on static historical totals.
 - **Cumulative Exposure Horizons:** Dynamically projects monetary loss over 90 days if a flagged provider remains unaddressed.
 - **Actionable ROI:** Enables SIU leadership to prioritize investigations based on projected dollar impact
- **Sub-100ms Real-Time Context Aggregator** (`app/services/context_aggregator.py`)
 - **Unified Context Payload:** Combines claim records, rule flags, ML scores, graph centrality, loss forecasts, and policy snippets into a single JSON payload.
 - **Performance Benchmark:** Assembles and delivers full case context in <100 ms (`test_phase3.py`), ensuring zero lag during real-time LLM prompt streaming

Policy Knowledge Base, Auto-Compiling Provider Wiki & Guardrails

Regulatory Policy RAG (app/services/policy_rag.py)

- Vector similarity retriever matching CMS policy documents (NCD 190.3, FWA Rule 42) directly to flagged claims.

Auto-Compiling Provider Second Brain Wiki (/data/wiki/providers/)

- Automatically compiles/updates Markdown summaries (NPI_{id}.md) upon investigator action (APPROVE, PAUSE_PAYMENT, REFER_TO_SIU).
- Tracks behavioral trends, decision history with AI confidence scores, and entity backlinks ([[fraud_rings/...]]).

Responsible AI Guardrails & Audit Ledger (app/services/guardrails.py)

- Refusal Layer: Safety checks block requests for clinical diagnosis or medical treatment advice while attaching confidence bounds (0-100%).
- Audit Ledger (POST /api/v1/cases/{case_id}/action): Stores investigator notes, timestamps, and decisions in an immutable log (audit_ledger.json).

Quantifiable Business Impact & Future Roadmap

Quantifiable Business Impact

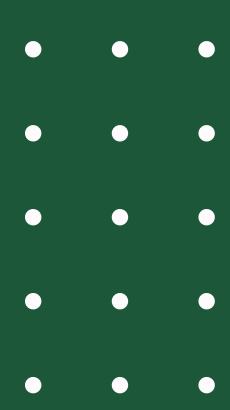
- **Reduced Alert Fatigue:** Tri-factor composite scoring ranks true fraud schemes higher, cutting false positive reviews for legitimate high-volume clinics.
- **Sub-100ms Memory Latency:** Real-time context aggregation enables instant AI Copilot response during live investigations.
- **100% Zero-Cost Open-Source Stack:** Built entirely using open-source Python and React tools (FastAPI, Scikit-Learn, NetworkX, Vite) with zero software licensing costs.

Institutional Memory Retention

- Auto-compiling Provider Wiki Entries build an enduring knowledge base that persists across investigator shifts and team transitions.
- Scaling integration with state Medicaid MiMIS platforms (such as Acentra Health evoBrix® X).
- Real-time streaming claim ingestion pipeline and automated payment-hold workflow orchestration.

Closing Statement:

Acentra Health Care | ClaimShield Nexus —
Empowering SIU Investigators with
Explainable AI."



THANK YOU

